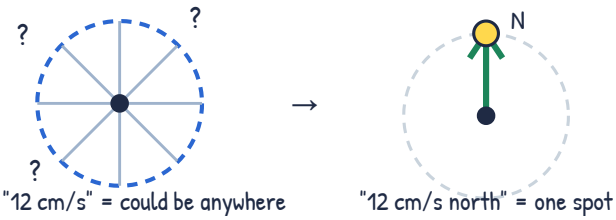
Motion test: read this on the way

Spelling doesn't count. Meaning does.

The 7 definitions (your teacher's exact words)

Motion	Change in position
Acceleration	The rate of change in velocity/speed
Reference Point	The location of the observer of motion
Velocity	Describing a moving object using speed and **direction**
Constant Speed	A speed that does not vary or change
Negative Acceleration	When an object's speed decreases (also called **deceleration**)
Positive Acceleration	When an object's speed increases

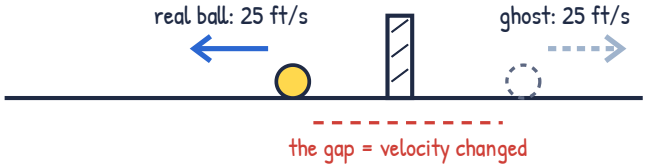
Why velocity needs a direction



"12 cm/s" = could be anywhere "12 cm/s north" = one spot

- Speed alone only tells **how far**. You need **which way** to know where it goes.
- **Direction** (north, left, toward the wall) is not **distance** (50 m).
- No direction given = **speed** (S). Has a direction = **velocity** (V).
  12 cm per sec → S. 3 m/sec North → V.

Same speed, new direction = new velocity



- The speedometer never moved, but the ghost (same velocity) and the real ball split apart.
- Bouncing back, turning a corner, going in a circle: the speed can stay the same but the **direction changes**, so the **velocity changes**.
- Walking 6 ft/s around corners: speed **was** constant, velocity **was not**.
- Ball bouncing off the wall: it **did not** have a constant velocity.

Careful: these trip people up

- Rate of change of **motion** (position) = **speed**. ~~acceleration~~
- Rate of change in **velocity/speed** = **acceleration**.
- Total distance ÷ total time = **average** speed.
  60 km in 1 h, then 60 km in 3 h: 120 ÷ 4 = **30 km/h**, not 40.
- Velocity is speed and **direction**. ~~distance~~
- Steady 60 mph in a straight line: acceleration is **zero**. Zero acceleration ≠ stopped.
- Slowing from 35 to 0 mph: velocity **did** change, it **did** accelerate, **negative** = **deceleration**.

Speed math: always show the work



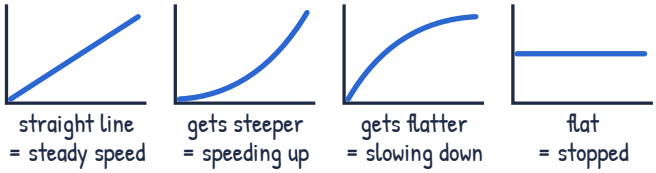
$s = d \div t$

$d = s \times t$ $t = d \div s$

Cover the letter you want.

- Write the formula, plug in, then answer **with units**:
  $S = d \div t = 150 \div 6 = 25$ ft/s
- Decimal trick: 180 ÷ 4.5 → double both → 360 ÷ 9 = **40 km/h**.
- More distance, same time → faster. More time, same distance → slower.

Distance-time graphs



- **Slope = speed.** Steeper line = faster.
- A flat line at 40 m: **not moving**, and **not at the start**. It's stopped 40 m away.

You know this. Read the question twice, then write the teacher's words.